

The Ephemeris

April 2014

Volume 25 Number 04 - The Official Publication of the San Jose Astronomical Association



Houge Park April Events

Friday, April 04

Public Star party: 8:30-10:30pm

Bear Creek Stables, Los Gatos
(in lieu of Houge Park this session)

Sunday, April 06

Solar observing: 2-4PM

Fix-It Day: 2-4PM

Saturday, April 12

Board of Directors Meeting: 6-7:30pm

General Meeting Social Time: 7:30-8pm

Guest Speaker; Derek Breit: 8pm

Topic: Occultations; Asteroidal and Lunar

Friday, April 18

Beginner Astronomy Class: 7:45-8:45pm

In-Town Star party (Houge): 8:45-10:45pm

*SJAA events are subject to cancellation due to weather.
Please visit website for up-to-date info.*

SJAA Contacts

President:	Rob Jaworski
Vice President:	Lee Hoglan
Treasurer:	Michael Packer
Secretary:	Teruo Utsumi
Director:	Greg Claytor
Director:	Dave Ittner
Director:	Ed Wong
Director:	Rich Neuschaefer
Director:	pending
Beginner Class:	pending
Fix-it Program:	Ed Wong
Imaging SIG:	Harsh Kaushikkar
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Speakers:	Teruo Utsumi

E-mails: <http://www.sjaa.net/contact-us/>

Exploratory Trip to Henry Coe State Park – Hunting Hollow Lot

By Ed Wong

This is my report of an exploratory trip to Henry Coe – Hunting Hollow Lot. I have been interested in exploring this parking lot as a potential site that club members could visit for dark sky viewing and imaging. I came across this site last January when I was out scouting for new dark sky sites and wanted to bring some members back with me for an Exploratory Trip. On Fri. March 21st Lee Hoglan, Gary Chock and I met at the site to explore and do some observing. First impression of the lot, it is big; 265 feet by 171 feet. For comparison, RCDO is 146 feet by 70 feet.

We setup our gear as the sun was going down. Lee brought the "son of the beast" scope, a 16 inch Dob, Gary had his Astro72 refractor and 20×80 Celestron Binoculars and I had my Oberwerk 28×110 Binoculars. As sun went down, we started to look at the objects in the west; the Pleiades, Orion Nebula, and Double Cluster. The Orion Nebula in the 16" dob was very bright and impressive but, so was the sky glow behind it due to the light dome in the west. I placed my DGM Optics NPB filter in the binoculars and the view was quite good, not as bright as the Dob but, very nice. With the NPB filters in the binoculars, the skyglow was diminished while the contrast and detail on the nebula really popped out.

Next we went on a galaxy hunt. In both the Dob and the 28×110 binos we had pretty good views of the Leo Triplet, M95, M96, M105, the Markarian Chain, M51, M101, M106, M108, M109, Owl Nebula, M81, M82, M64. Gary was having some nice views of the bigger star clusters like the Hyades, Beehive and Mel 111.

We ended up wrapping up about 10:30pm and all agreed it was one of the best nights of observing we have had in a long time.

As far as the site qualities go: The Good – This site is big, all of the regular SJAA members who come out to view on a regular basis could fit in that lot with room to spare. There is no lock or gate at the entrance so it's 7/24 hour

access. The use fee is \$6 per person and is deposited right on site. The site is an easy drive, within 1 hour from Houge Park and is 10 minutes from Mendoza Ranch. There were very few cars on the road I think mostly because the access road dead ends shortly after the site entrance. The site is pretty dark, SQM meter reading of 21.1 around 9:30pm. There is also camping if you choose to stay overnight.

The Bad – The is no lock or gate at the entrance so it's 7/24 hour access which means we have no control over other cars driving in while we are viewing. Horizons not that low; from the center of the lot, the south is about 30 degrees, east and west about 20 degrees and north is about 15 degrees. The elevation is about 600 feet so there could be fog on some nights.

The Ugly – Possible that night hikers could pull into the lot, happened to us. They had some lights on for about 15mins as they were gathering their stuff to hike into the park. For this reason, the site seems more suited to viewing than imaging.

My overall impressions of the site: I think this is a good site for viewing. It is the darkest of the sites that club members currently go to (RCDO, Mendoza Ranch). It's good to know that this site is available for use on a 7/24 basis. While it's not perfect, neither are the other sites that we have access to. I think overall the good qualities do outweigh the bad on this site.

The GPS location of the site is:
37°04'33.6"N 121°27'59.4"W

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SJAA AUCTION - March 16, 2014



GOING ONCE



SOLD!!!



GOING TWICE



Adventures of a Stargazer

From Lee Hoglan

Because of the nature of what we do, it's no surprise amateur astronomers have some unexpected adventures (MISADVENTURES) while heading out to stargaze, set up a telescope or build an observatory. Some experiences are funny and some not.

When I was in college in the 70's I had a 16" Sonotube Newtonian on a trailer. Boy, did I get some strange looks driving down the highway. Very few knew what it was. Most thought it was some kind of artillery launcher, some thought I was part of a traveling circus and the rest wondered Why I was hauling around a hot water heater.

Then there was the time outside of Lake Almanor when I was out on a dark country road late at night. I had a big pair of binoculars in the car. I was driving kind of slowly looking for a side road to pull off and do some stargazing. I didn't think too much of it when a car with two guys pulled up behind. I drove on looking for a clearing. Well, I guess the guys thought anybody driving that slowly was either drunk or up to no good because it wasn't long before red and blue lights started flashing behind me. After pulling over and explaining my intention, and giving the local sheriffs an astronomy lesson, I proceeded to a nice open field by a canal where my big binoculars made the Andromeda galaxy look like a sparkling Frisbee.

Another adventure I could have done without. On the open ranch land by Mercey Hot Springs my observing buddy Bob and I were building a concrete observatory building to house two telescopes. The construction workers related a strange tale of goings-on. It seems the Fresno County Sheriff's department sent out half the force including a swirling helicopter to storm this strange new building. They forced their way through the burglarproof doors and got in. I think they were expecting to find an illicit meth lab or a UFO base or at least Weapons of Mass Destruction. Instead of finding Osama bin Laden's hideout they must have been disappointed to find couple of old coats, some Gatorade and an old 12" Meade. Well at least they left a souvenir. It seems in their rush to get in and out they left a pair of sheriff storm trooper boots. I never did get any explanation from the cops.

And now for the animal story.

Last summer I was on my way to the observatory at DARC looking forward to a night of finding some Caldwell dim fuzzies. I was driving the winding road to the Panoche wilderness area with my father, when I came around a corner and saw what appeared like a giant white box with a horse's head sticking out by the side of the road.,. After a dazed moment I realized I had come upon an accident that had just happened in pitch darkness. I soon found out a woman had suffered a blowout while hauling her horse trailer carrying two horses. The horses were bloody and kicking to get out of the flipped over trailer. Fortunately I had some tools in my Explorer and with the help of a couple of passersby and my hacksaw and hammer and a couple of hours of cutting and banging, we managed to extricate scared equines from the upside down transport. Well, after leaving the scene exhausted and covered in blood, I proceeded on to the observatory. After that intense experience, the idea of gazing at twinkling stars in my telescope seemed to lose its appeal. I just sat and looked up at the

dark sky the rest of the night.

There're some teeth in this next story.

My dad is getting up there in age, and he's never been much of a stargazer, but he likes the ride. So I take him to the observatory when I can. On a recent occasion there were about a dozen observers around the observatory enjoying a picnic. My father decided to remove his false teeth and stash them in a paper cup. That was a very bad idea. Of course the cup got tossed out and I spent the entire next morning sifting through old wrappers and paper plates and coffee grounds looking for my dad's fake chompers. I did find them in a black garbage bag in the back of my friend's pickup truck. Maybe he should get implants.

So much for the misadventures in astronomy. I think I'll take up Chess.

SJAA Yosemite Star Party

From Jim Van Nuland

The annual SJAA Yosemite star party will be held on July 25 and 26, at Glacier Point in Yosemite National Park. Up to 30 people will be given free admission and camping, in exchange for two public events on Friday and Saturday evenings. In what time is left, we can be tourists. We are expected to have at least one scope per two people, and to attend both star parties, not just Friday or Saturday. For these reasons, this is probably not suitable for a family camping trip. The camping is rough by modern standards: no dining room, no showers, no hot water. Read about it on the SJAA Yosemite page <<http://www.sjaa.net/school/yosemite.htm>> and the Yosemite FAQ page at <<http://www.sjaa.net/school/yosefaq.htm>>. Then contact me with remaining questions. That first page also has sun and moon rise and set times. The luck of the draw went with us this year -- this weekend is the 2nd choice on our request list. The moon is new on Saturday! If you can tolerate the limitations, tell me the number of people you'll have, and the number of scopes that will be set up for the public. E-mail me at <jvn@sjpc.org>. Priority is given to SJAA members. Clear Skies!

2014 Golden State Star Party

June 25-29, 2014

From Michael Packer

Early Registration for the Golden State Star Party closed at end of March. A dozen or more SJAA members were there last year and over 400 Astronomers from up and down the coast. Great BBQ! And last year we had some of the darkest skies on record. Did we mention great BBQ! We'll see you there June 25-29, Frosty Acres Ranch for another fun and memorable GSSP experience!

School Star Parties



The San Jose Astronomical Association conducts evening observing sessions (commonly called "star parties") for schools in mid-Santa Clara County, generally from Sunnyvale to Fremont to Morgan Hill.

The co-coordinator for SJAA is Jim Van Nuland

Big-Bang Theory: New Evidence

Some 13.8 billion years ago, just before the Big Bang, the enormous, galaxy-filled universe we know today was contained inside a tiny, dense, extremely hot point. Suddenly, it began rapidly expanding faster than the speed of light in a cataclysmic explosion. The universe grew from a subatomic size to that of a golf ball in an incomprehensibly short fraction of a second. This earliest instant of expansion, known as cosmic inflation, explains why the universe is relatively uniform (galaxies that formed as the universe cooled, for instance, appear to be scattered evenly as far as the telescope can see) and also explains the seeds of density that gave rise to the universe's structure.

It's a nice story, but for decades after physicists proposed it, our evidence for it has been limited. Our chief means of studying the Big Bang—the faint radiation left over from the explosion called the cosmic microwave background (CMB)—dates to about 380,000 years afterward, instead of the moment itself.

A significant new piece of evidence emerged March 17 2014, when a group of scientists led by astronomer John Kovac of the Harvard-Smithsonian Center for Astrophysics announced that they've found indirect evidence of gravitational waves—minute distortions in the universe's gravitational field—that were unleashed during inflation, a tiny fraction of a second after the Big Bang. If the finding is correct, the waves serve as a confirmation of inflation.

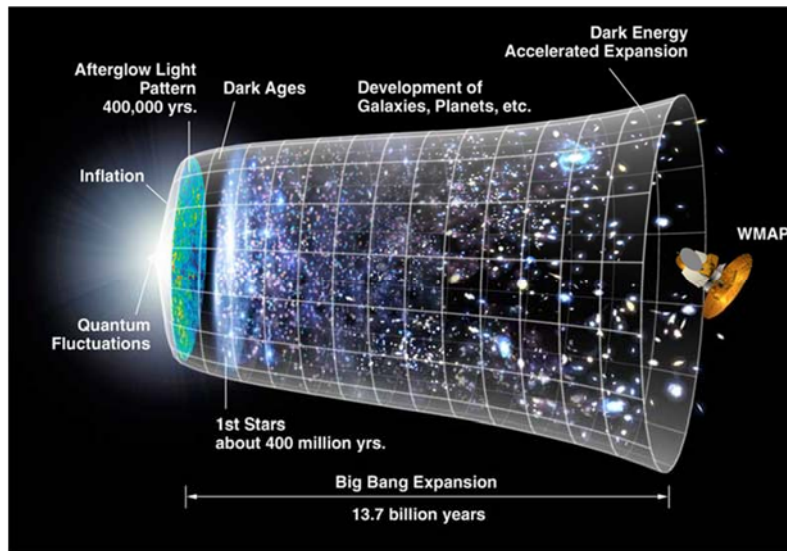
"Inflation is the 'bang' of the Big Bang," says theoretical physicist Alan Guth, who proposed the theory of cosmic inflation in 1979. "It's the mechanism that caused the universe to enter this period of gigantic expansion."

Several physicists who were not involved in the research have had the chance to assess the raw data, and they agree with the analysis. "It's very, very likely that this is real," says Avi Loeb, a theoretical physicist at the Harvard-Smithsonian Center, noting that the researchers spent three years analyzing the data to eliminate any chance of an error.

Robert W. Wilson, who shared the 1978 Nobel Prize in physics for his discovery of the cosmic microwave background, agrees, and believes that if confirmed, the work is almost certain to garner a Nobel Prize. Loeb says the finding would be one of the most important physics discoveries of the past 15 years—bigger than the discovery of the Higgs Boson.

During inflation, shown at far left, the universe expanded by many orders of magnitude in size in a fraction of a second.

The presence of detectable gravitational waves—often referred to as "ripples in the fabric of spacetime"—is predicted by the theory of inflation. Pre-existing fluctuations in the strength of gravity on a microscopic scale, Guth says, would have been stretched out by inflation, producing macroscopic waves.



(Image via NASA)

The exact nature of the waves depends on the precise moment inflation occurred. "This detection is not only indicating that inflation took place," Loeb says, "but also telling us when it took place": 10^{-34} (a decimal point followed by 33 zeroes and then a one) seconds after the start of the Big Bang.

The research group, which also included Clement Pryke of the University of Minnesota, Jamie Bock of Caltech and Chao-Lin Kuo of Stanford, didn't find gravitational waves themselves, but rather indirect evidence of them, in the form of a special pattern of polarization caused by the waves in

the cosmic microwave background. "Our team hunted for a special type of polarization called B-modes, which represents a twisting or curl pattern in the polarized orientations of the ancient light," Bock said in a press statement.

The researchers collected this data using the BICEP2 telescope, stationed in Antarctica, where cold, dry air limits interference from the Earth's atmosphere on the faint cosmic microwave background signal. BICEP2 is one of a suite of identical telescopes looking for this signature, called the Keck Array. There's also the adjacent South Pole Telescope, which reported data that indicated the presence of B-mode polarization in the CMB last summer.

That instrument, however, wasn't designed to detect polarization at the scale produced by gravitational waves, so it likely resulted instead from the interference of distant galaxies that the CMB passed through before reaching Earth.

The BICEP-2 Telescope (the white dish at right), along with the South Pole Telescope (at left).

It's still not entirely clear that the BICEP2 team has detected B-mode polarization that is in fact definitive proof of gravitational waves. Further confirmation will need to come from data collected by the European Space Agency's Planck Satellite (which is observing the cosmic microwave background at a much wider angle), due to be released at the end of the summer.

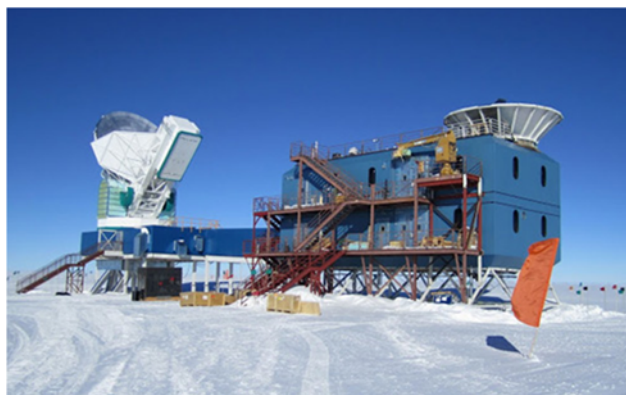
If true, though, the find would go a long way toward ratifying the theory of inflation. "The presence of this polarization, induced by the gravity waves, is the last big thing predicted by inflation," Wilson says. "It gives you ever-more confidence that this really is the correct scenario."

It would also reflect something truly astounding: the oldest evidence we have of absolutely anything.

"You can't use cosmic microwave background to figure out what happened in the early, early universe,"

Loeb says. For the first 380,000 years, the electromagnetic waves that make up the CMB couldn't pass freely through space. "If we can look at gravitational waves, we can go all the way back to almost the very beginning."

Credit: Smithsonian



(Image via BICEP-2 Project)

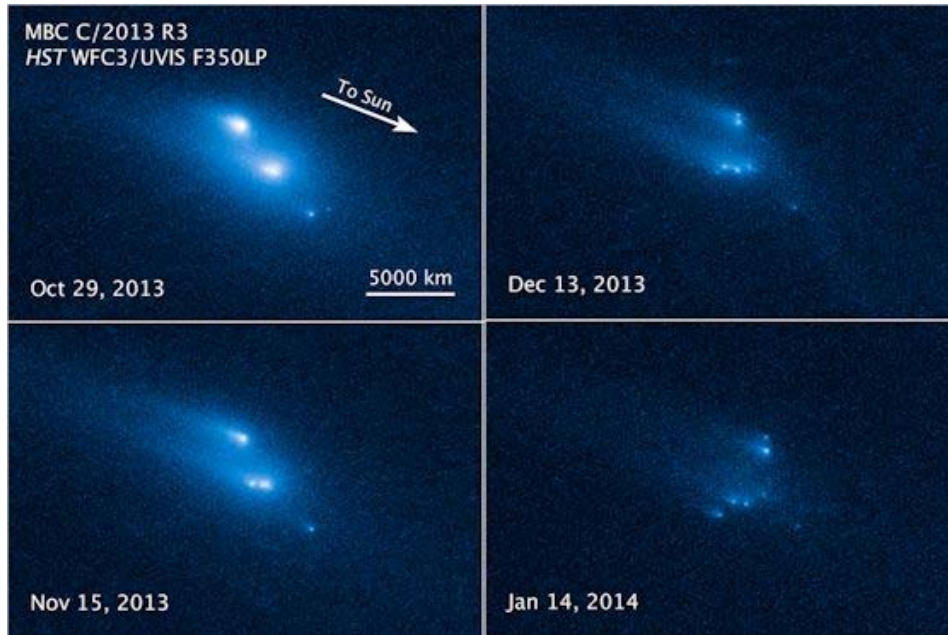
Hubble Witnesses Asteroid's Mysterious Disintegration

NASA's Hubble Space Telescope has recorded the never-before-seen break-up of an asteroid into as many as 10 smaller pieces. Fragile comets, comprised of ice and dust, have been seen falling apart as they approach the sun, but nothing like this has ever before been observed in the asteroid belt.

"This is a rock, and seeing it fall apart before our eyes is pretty amazing," said David Jewitt of the University of California at Los Angeles, who led the astronomical forensic investigation.

The crumbling asteroid, designated P/2013 R3, was first noticed as an unusual, fuzzy-looking object by the Catalina and Pan STARRS sky surveys on Sept. 15, 2013. A follow-up observation on October 1 with the W. M. Keck Observatory on the summit of Mauna Kea, a dormant volcano on the island of Hawaii, revealed three bodies moving together in an envelope of dust nearly the diameter of Earth.

"The Keck Observatory showed us this thing was worth looking at with Hubble," Jewitt said. "With its superior resolution, space telescope observations soon showed there were really 10 embedded objects, each with comet-like dust tails. The four largest rocky fragments are up to 400 yards in diameter, about four times the length of a football field." Hubble data showed the fragments drifting away from each other at a leisurely one mph. The asteroid began coming apart early last year, but new pieces continue to reveal themselves, as proved in the most recent images.



This series of Hubble Space Telescope images reveals the breakup of an asteroid over a period of several months starting in late 2013. The largest fragments are up to 180 meters (200 yards) in radius.

It is unlikely the asteroid is disintegrating because of a collision with another asteroid, which would have been instantaneous and violent by comparison to what has been observed. Debris from such a high-velocity smashup would also be expected to travel much faster than observed. Nor is the asteroid coming unglued due to the pressure of interior ices warming and vaporizing. This leaves a scenario in which the asteroid is disintegrating due to a subtle effect of sunlight, which causes the rotation rate of the asteroid to gradually increase. Eventually, its component pieces -- like grapes on a stem -- succumb to centrifugal force and gently pull apart. The possibility of disruption in this manner has been

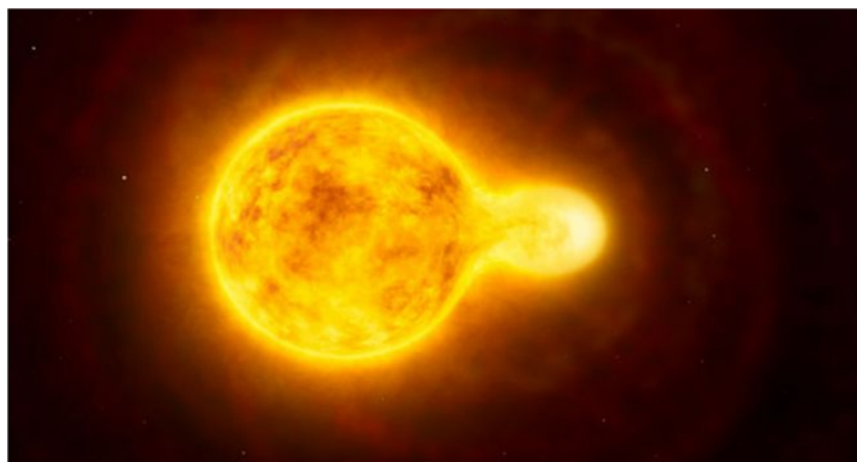
discussed by scientists for several years, but never reliably observed. For this scenario to occur, P/2013 R3 must have a weak, fractured interior -- probably as the result of numerous non-destructive collisions with other asteroids. Most small asteroids are thought to have been severely damaged in this way. P/2013 R3 is likely the byproduct of just such a collision sometime in the last billion years. With the previous discovery of an active asteroid [spouting six](#)

[tails](#), named P/2013 P5, astronomers are finding more evidence the pressure of sunlight may be

the primary force causing the disintegration of small asteroids -- less than a mile across-- in our solar system. The asteroid's remnant debris, weighing about 200,000 tons, will in the future provide a rich source of meteoroids. Most will eventually plunge into the sun, but a small fraction of the debris may one day blaze across our skies as meteors.

Credit: Tony Phillips, NASA

This artist's impression shows the yellow hypergiant star HR5171. This is a very rare type of star with only a dozen known in our galaxy.



Largest yellow star ever seen revealed in new light



Kid Spot Jokes:

- **What does an astronaut do when he gets angry?**
He blasts off.
- **What do you call a loony spaceman?**
An astronut!

Kid Spot Quiz:

1. What is the name of the first satellite sent into space?
2. What is the name of NASA's most famous space telescope?

Kid Spot Night Sky Challenge: March 2014

See if you can spot the following objects in the sky:

- ⇒ Jupiter – All month
- ⇒ Mars – Will be closest to Earth and will be very bright!
- ⇒ Total lunar eclipse – April 15
- ⇒ Spica – in the constellation Virgo

<http://skyandtelescope.com/observing/atagance>



Photo Credit: Wikipedia

Constellations

Leo—The Lion

Leo /'li:os/ is one of the constellations of the zodiac, lying between Cancer to the west and Virgo to the east. Its name is Latin for lion, and to the ancient Greeks represented the Nemean Lion killed by the mythical Greek hero Heracles (known to the ancient Romans as Hercules) as one of his twelve labors.

Its symbol is (♌). Leo is one of the 48 constellations described by the 2nd century astronomer Ptolemy and Leo remains one of the 88 modern constellations today, and one of the most easily recognizable due to its many bright stars and a distinctive shape that is reminiscent of the crouching lion it depicts.

The lion's mane and shoulders also form an asterism known as "**the Sickle**", which to modern observers may resemble a backwards "**question mark**."

Leo contains many bright stars, many of which were individually identified by the ancients. There are four stars of first or second magnitude, with render this constellation especially prominent:

Regulus, designated Alpha Leonis, is a blue-white main-sequence star of magnitude 1.34, 77.5 light-years from Earth. It is a double star divisible in binoculars, with a secondary of magnitude 7.7. Its traditional name (Regulus) means "the little king".

Beta Leonis, called **Denebola**, is at the opposite end of the constellation to Regulus. It is a blue-white star of magnitude 2.23, 36 light-years from Earth. The name Denebola means "the lion's tail".



Algieba, Gamma Leonis, is a binary star with a third optical component; the primary and secondary are divisible in small telescopes and the tertiary is visible in binoculars.

Leo contains many bright galaxies; Messier 65, Messier 66, Messier 95, Messier 96, Messier 105, and NGC 3628 are the most famous, the first two being part of the Leo Triplet. The Leo Ring, a cloud of hydrogen and helium gas, is found in orbit of two galaxies found within this constellation. Leo was one of the earliest recognized constellations, with archaeological evidence that the Mesopotamians had a similar constellation as early as 4000 BCE. The Persians called Leo *Ser* or *Shir*, the Turks, *Artan*; the Syrians, *Aryo*; the Jews, *Arye*; the Indians, *Simha*, all meaning "lion".

Some mythologists believe that in Sumeria, Leo represented the monster

Khumbaba, who was killed by Gilgamesh.

In Babylonian astronomy, the constellation was called UR.GU.LA, the "Great Lion"; the bright star Regulus was known as "the star that stands at the Lion's breast." Regulus also had distinctly regal associations, as it was known as the King Star.

Source: Wikipedia

Kid Spot Quiz Answers:

- 1) Sputnik
- 2) Hubble

SJAA Library

From Sukhada Palav

This last month SJAA library underwent some infrastructural makeover and is almost ready for you! Please visit the library page for more details - <http://www.sjaa.net/sjaa-library/>.

Please also take a look at our Wish list for kids' books posted on the SJAA webpage. SJAA library will gladly accept donations towards any of these books. Your feedback on our collection or specific books is also welcome! I can be reached for any questions or comments at:

librarian.sjaa@gmail.com

Advanced Loaner Telescope Program

From Dave Ittner

The purpose of this program is for SJAA members to evaluate equipment they are considering purchasing or are just curious about. Check out the growing list of equipment on the web page. Please note that certain items have restrictions or special conditions that must be met.

If you are an SJAA member and an experienced observer or have been through the SJAA **QuickSTART** program please contact Dave Ittner to request a particular item. Please consider donating unused equipment.

Fix-It Program

From Ed Wong

If you plan on bringing something to the FixIt session inside Building 1, though it's not required, we ask that you check out the FixIt page on the SJAA website and fill out the form so we know what to expect.

Here's where to find it all:

<http://www.sjaa.net/fixit>

From the Board of Directors

General Notices

- SJAA is signed up for Cupertino's 6th Annual Earth Day Festival, Saturday, April 5th, 2014 from 11am - 3pm
- SJAA is signed up for Astronomy Day and SJ MLK Public Library Saturday May 10 2014 from 11am - 3pm.
- The annual SJAA Yosemite star party will be held on July 25 and 26, 2014, at Glacier Point in Yosemite National Park. See the SJAA webpage for more information.
- The Golden State Star Party (GSSP)

will take place June 25-29, 2014. See Michael Packer or the SJAA BLOG for more details.

- There is currently one (1) board seat open that needs to be filled. See Lee Hoglan or any board member if you are interested.

Announcements

- **The 34th annual SJAA Auction is over. Thank you to all who participated!**
- Volunteers are needed to help check in/out books at various Houge Park events. Contact club librarian Sukhada Palav if interested.
- The next Public Star Party will be held Friday, April 4th at Bear Creek Stables in lieu of Houge Park.

Board Meeting Excerpts March 15, 2014

In attendance

Rob Jaworski, Lee Hoglan, Ed Wong, Greg Claytor, Dave Ittner, Teruo Utsumi, Rich Neuschaefer, Michael Packer

Officers

Selection and Approval of 2014 Officers:

President; Rob Jaworski
Vice President; Lee Hoglan
Treasurer; Michael Packer
Secretary; Teruo Utsumi

New Alternate Viewing Site

The Friday, April 4th In-Town Start Party will be held at Bear Creek Stables.

Library

Sukhada Palav will generate an announcement regarding a new book wish list for the Club library.

Total Lunar Eclipse Party April 14-15

Rich Neuschaefer volunteered to run the lunar eclipse event which will start Monday night and end early Tuesday morning.

SJAA Auction 2014

From Rob Jaworski

The 34th annual SJAA Auction was a smashing success! With an energized crowd, lots of humor and good spirits during bidding, and lots of items being sold for

good amounts, this annual event again proved its worth in the value that it brings to the amateur astronomy community throughout the bay area, and throughout Northern California.

There will be more write ups on the SJAA blog, but preliminary numbers show that there were about fifty bidders, ten sellers, well over a hundred individual lots being made available, and over \$6700 worth of transactions. Including the auction sale of donated telescopes and other gear, initial estimates indicate the club netted over \$2600 as a result of the event.

The crowd was in great spirits throughout the day. There was pizza, crock pot fire-house chili, sweet desserts, soda pop and other edibles available. The event started on time at noon and ended at around 4PM, just as planned. Many people commented how smoothly it ran despite all the new folks working the critical functions, not least to mention was Greg Claytor in his premiere performance as auctioneer. Greg kept the crowd entertained, while working to secure the highest price for the seller, yet maintained good deals for the bargain hunters in the crowd.

Thank you to all who came and were a part of this signature SJAA event, sellers, bidders and observers all!

Dark Sky Events

Apr-May 2014

Friday, April 04

Public Star party: 8:30-10:30pm
Bear Creek Stables, Los Gatos

April 19

Starry Nights Star Party: 9-11pm
Rancho Canada del Oro (RCDO)

Saturday, April 26

Henry Coe State Park
Mendoza Ranch

Saturday, May 17

Starry Nights Star Party: 9-11pm
Rancho Canada del Oro (RCDO)

Saturday, May 31

Henry Coe State Park
Mendoza Ranch

SJAA Ephemeris newsletter of the San Jose Astronomical Association, is published monthly

Articles for publication should be submitted by the 20th of the previous month.

San Jose Astronomical Association
P.O. Box 28243
San Jose, CA 95159-8243
<http://www.sjaa.net/contact>

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here

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San Jose, CA 95159-8243

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San Jose Astronomical Association Membership Form

P.O. Box 28243 San Jose, CA 95159-8243

☐ **New** ☐ **Renewal** (Name only if no corrections)

Membership Type:

- ☐ Regular — \$20
☐ Regular with Sky & Telescope — \$53
☐ Junior (under 18) — \$10
☐ Junior with Sky & Telescope — \$43

Subscribing to Sky & Telescope magazine through the SJAA saves you \$5 off the regular rate. (S&T will not accept multi-year subscriptions through the club program. Allow 2 months lead time.)

☐ **I prefer to get the Ephemeris newsletter in print form (Add \$10 to the dues listed on the left). The newsletter is always available online at:**

<http://www.sjaa.net/sjaa-newsletter-ephemeris/>

Questions? Send e-mail to
sjaamemberships@gmail.com

Bring this form to any SJAA Meeting or send to the address (above). Make checks payable to "SJAA", or join/renew at <http://www.sjaa.net/join-the-sjaa/>

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